

1 Does contrastive decoding act where the hallucination is decided?

Context. Contrastive decoding (CD) subtracts a degraded amateur branch from the expert to push the model away from hallucinated answers. If this works as intended, then at the token where the model commits to an answer the subtraction should be *selective*: it should lower the score of answers that are actually hallucinations while leaving correct answers unchanged. A shift that moves every answer by the same amount, regardless of whether it is a hallucination, is not a correction but a uniform translation of the decision. We test which of the two it is by looking inside the model, at the representation it uses to decide.

Methodology. We restrict the test to the questions the model answers “Yes,” the only answers on which a hallucination can occur, and split them into hallucinations $\mathcal{H}$ (object absent, model says Yes) and correct calls $\mathcal{T}$ (object present, model says Yes). To read a decision off a layer we use the model’s own head (the logit lens): apply the final RMSNorm and unembedding to the layer’s hidden state, restrict to the “Yes”/“No” tokens, and take the margin $m_\ell = \max_{\text{Yes}} z_\ell - \max_{\text{No}} z_\ell$, positive when the model would answer “Yes.” Running this on the clean (expert) and noised (amateur) passes gives m_ℓ^E, m_ℓ^A ; since CD forms $(1 + \alpha)z^E - \alpha z^A$ and the margin is linear in the logits,

$$m_\ell^{CD} = m_\ell^E + \alpha \Delta_\ell, \quad \Delta_\ell = m_\ell^E - m_\ell^A, \quad (1)$$

so Δ_ℓ is one number per sample per layer: the log-odds CD adds to the decision, negative when it suppresses “Yes” (the correcting direction on a hallucination) and positive when it reinforces it. We also fit a linear probe (logistic regression, standardized features, 5-fold CV, out-of-fold probabilities) on the hidden state at each layer and report its AUC.

Experiment 1 (direct test). Layer-wise analysis is the natural lens here because the final answer is nothing more than a readout of a hidden state: the decision is formed inside the stack, at whatever depth the hallucination signal becomes linearly present, and only then projected to the “Yes”/“No” tokens. If CD genuinely corrected the hallucination, it would have to engage *that* representation, so its effect should be largest at, or at least track, the layer where the hallucination is decided. A correction that lands where the deciding information is not yet present, or that ignores that depth entirely, is acting on something other than the mechanism that produced the error. The intuitive test is therefore to ask where, across depth, the hallucination is decided and whether that is where CD acts. We plot, per layer, how decodable the hallucination is (a probe over the absent samples predicting the false “Yes”) against CD’s mean effect $\bar{\Delta}_\ell$ on $\mathcal{H}$ (Figure 1). This is the natural first cut, but two problems keep it from settling the question. First, the residual stream is additive: a shift written at any layer propagates to the output, so a mismatch between where the hallucination is decoded and where $\bar{\Delta}_\ell$ peaks does not by itself prove CD acts in the wrong place. Second, $\bar{\Delta}_\ell$ is a magnitude and a group mean; a large average shift says nothing about whether CD *distinguishes* hallucinated from correct answers, which is what a correction must do.

Experiment 2 (selectivity). To remove both problems we test, at every layer, whether Δ_ℓ carries hallucination-specific information rather than where it is largest. Since $\mathcal{H}$ is absent and $\mathcal{T}$ present, telling them apart is exactly object presence, so the probe of presence (Probe A) is the matched baseline for what CD would need. We compute four layer-wise quantities on $\mathcal{H} \cup \mathcal{T}$ (Figure 2): (i) a *selectivity* AUC, whether $-\Delta_\ell$ ranks $\mathcal{H}$ above $\mathcal{T}$ for suppression (0.5 if blind); (ii) whether the amateur branch even forms the presence signal, the AUC of m_ℓ^A vs. the clean m_ℓ^E ; (iii) the fraction of $\mathcal{H}$ with $\Delta_\ell > 0$; and (iv) R_ℓ^2 , the share of the variance of Δ_ℓ explained by hallucination status. Because these are layer-resolved and about information content, a null result at *every* depth closes the additivity loophole: there is no layer, early or late, at which a selective correction could hide.

Experimental details. We run POPE-COCO (9000 questions) on LLaVA-1.5-7B. For each question we take two single forward passes at $T = 1$, a clean pass and a noised pass corrupted by the method’s own `add_diffusion_noise(·, 900)`, and store the logit-lens Yes/No margins at the final prompt-token position for all 33 layers. Of the 3710 questions answered “Yes,” $|\mathcal{H}| = 257$ are hallucinations and $|\mathcal{T}| = 3453$ correct positives. Probe A is fit on all samples; the hallucination-decodability probe of Experiment 1 (Probe B) on the absent samples. Extraction ran on a single NVIDIA L4; the probes and analyses are offline on CPU. We use $\alpha = 1$, and verified the lens reproduces the true final logits to a mean absolute error of 1.7×10^{-3} log-odds.

Results. Experiment 1 (Figure 1) shows the hallucination is decodable from layer 12 (probe AUC $\rightarrow 1.0$), while CD’s effect is near zero until then, peaks at layer 19 ($\bar{\Delta} = +17.9$), and decays to $+3.32$ at the readout. Two features stand out: CD’s effect is offset several layers past where the hallucination is settled, and it is *positive* at every layer, i.e. it reinforces the false “Yes” rather than suppressing it. On its own, though, this is only suggestive: by the additivity argument the depth offset need not imply CD acts in the wrong place, and a positive mean does not establish that CD treats hallucinated and correct answers alike.

Experiment 2 (Figure 2, Table 1) settles both. Object presence is decodable from layer 8 (Probe A AUC crosses 0.95, rising to 0.99; even the 1-D margin separates $\mathcal{H}$ from $\mathcal{T}$ at AUC 0.86), so the information a selective CD needs is present. Yet CD’s selectivity never leaves $[0.40, 0.60]$ at any depth and is 0.52 at the readout; its shift reinforces the false “Yes” for 96.5% of hallucinations at the readout, with the fraction pushed toward “Yes” locking near 0.95 from layer 17; and hallucination status explains at most 0.7% of the variance of Δ_ℓ anywhere, including layer 19 where the raw shift is largest. The amateur branch keeps a weaker but real presence signal (AUC 0.78 vs. the clean 0.94), so CD is not blind for lack of a signal; it simply does not encode presence at any depth. Because the selectivity is null at every layer, the additivity loophole is closed. Two limits remain: this is the single forced yes/no token, not free-form generation, and it isolates the contrastive term $m^E + \alpha\Delta$ (the decoder also applies a plausibility mask); changing α scales Δ but cannot flip its sign, so no α recovers suppression once $\Delta > 0$ for 96.5% of cases.

What we infer. From layer 8 the model carries a clean linear representation of object presence, exactly what separates a hallucinated “Yes” from a correct one, but CD’s shift never uses it: its selectivity sits at chance, it explains under 1% of the shift’s variance, and where it acts most strongly it pushes hallucinations further toward “Yes.” Contrastive decoding therefore does not correct hallucinations at the point of decision; it applies a uniform, one-sided translation of the Yes-vs-No margin, blind to whether an answer is a hallucination, so any measured reduction is a threshold effect rather than a correction of the mechanism that produces it.

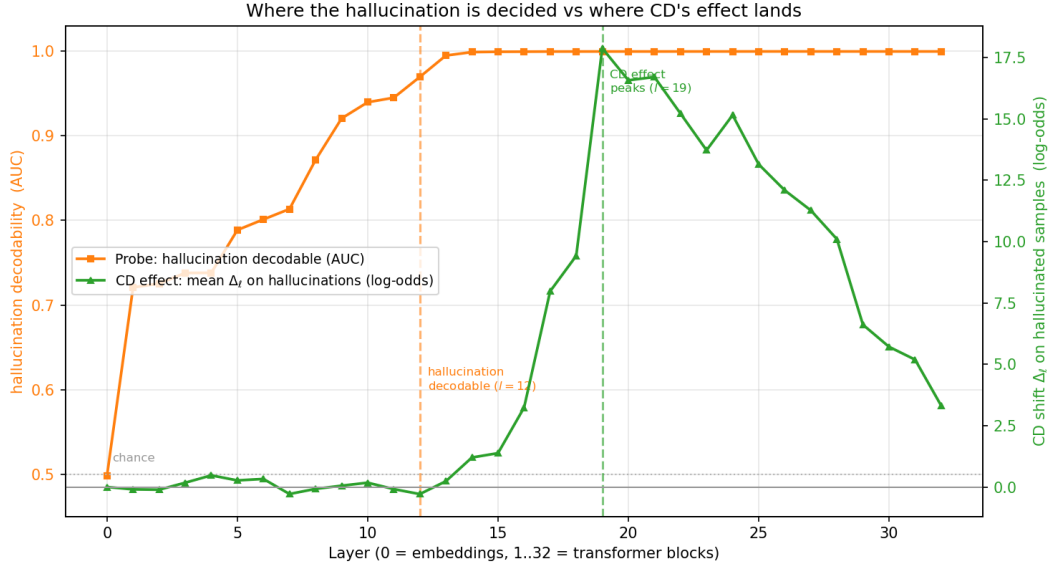


Figure 1: Experiment 1 (direct test). Hallucination decodability (orange, left) is complete by layer 12, while CD’s mean effect $\overline{\Delta}_\ell$ on $\mathcal{H}$ (green, right, log-odds) is near zero until then, peaks at layer 19, and stays positive throughout, i.e. CD reinforces the false “Yes.” Suggestive but not conclusive: additivity means the depth offset is not decisive, and a group-mean magnitude does not establish selectivity.

Layer-wise test: CD's shift never separates hallucinated from correct Yes-answers, at any depth

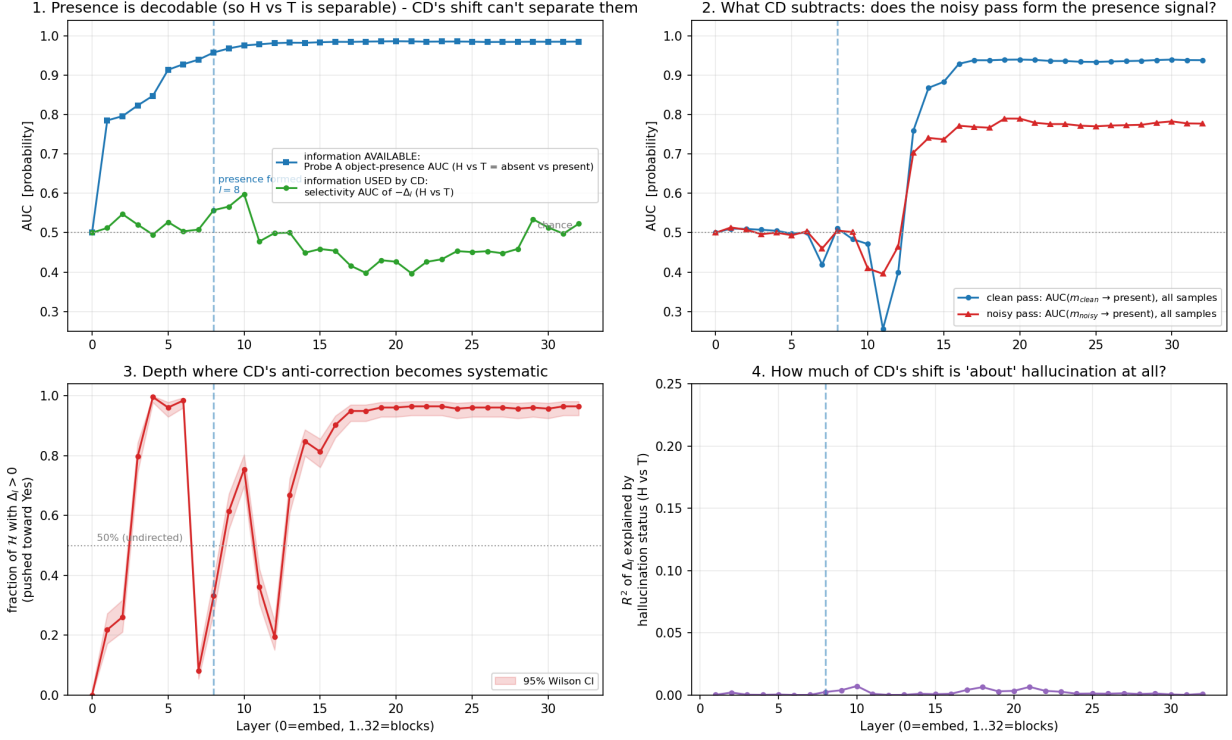


Figure 2: Experiment 2 (selectivity). (1) Presence (the $\mathcal{H}$ -vs- $\mathcal{T}$ distinction) is decodable from layer 8 (Probe A, blue), yet CD’s selectivity (green) stays at chance. (2) The amateur branch’s presence signal is degraded (AUC 0.78 vs. 0.94) but not absent. (3) The fraction of $\mathcal{H}$ pushed toward “Yes” locks near 0.95 from layer 17 (95% Wilson interval). (4) Hallucination status explains under 1% of the variance of Δ_ℓ at all depths.

Quantity	Value
Answered “Yes”: hallucinated $\mathcal{H}$ / correct $\mathcal{T}$	257 / 3453
Presence decodable (Probe A AUC ≥ 0.95) from layer	8
Presence AUC: Probe A / in-population margin	0.99 / 0.86
Selectivity AUC of $-\Delta_\ell$: range / readout	[0.40, 0.60] / 0.52
Presence AUC at readout: clean / noisy branch	0.94 / 0.78
Fraction of $\mathcal{H}$ with $\Delta > 0$ at readout	0.965
Variance explained R_ℓ^2 : max / readout	0.007 / 0.0009
Mean shift $\bar{\Delta}$ at readout: $\mathcal{H}$ / all	+3.32 / +1.23

Table 1: Numerical results ($\alpha = 1$), from the per-sample logit-lens margins over all 9000 questions.